



OAKLAND COMMUNITY COLLEGE*

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140

Studio 5000 FUNCTION BLOCK LABS

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IMPORTANT FOR GRADE



For each lab, the **named Quick Watch** must be created and displayed during program execution for credit.

See graphic Completed Watch Window for required tags.

TABLE OF CONTENTS

Studio 5000 FUNCTION BLOCK LABS

LAB: SS1.....	1
LAB: WASHER STATION TIMER	10
LAB: TIMED PROCESS OUTPUT	17

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LAB: SS1**PROCESS MODEL:** SS1 {last name} {first name}**OBJECTIVE:** Create a simple Stop-Start circuit with function block instructions.

START - STOP circuit for a single motor control with ON - OFF indicators.

INPUT:

Address	Controller Tag	Descriptions
Local:3:I.Data.0	STOP	Pushbutton Momentary N.C.
Local:3:I.Data.1	START	Pushbutton Momentary N.O.

OUTPUT:

Address	Controller Tag	Descriptions
Local:4:O.Data.0	PL1_OFF	Pilot Light 1 OFF (Red)
Local:4:O.Data.1	PL2_ON	Pilot Light 1 ON (Green)
Local:4:O.Data.2	MOTOR	Motor Control

FUNCTION BLOCKS:

Function Block Tags are created as the instructions are added.

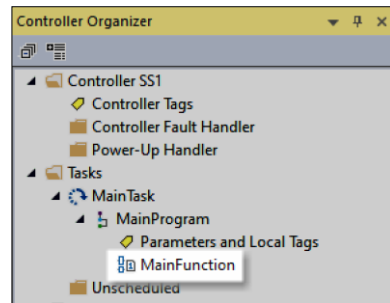
Param/Local Tag	Descriptions
BAND_01	AND for STOP Pushbutton Holding Circuit
BOR_01	OR for START Pushbutton Holding Circuit
BNOT_01	Invert for PL1_OFF Pilot Light

PROGRAM LOCATION:**Studio 5000** Folder**Course Day/Time** Folder**Languages** Folder**Function Block** Folder**FILE NAME:** Function_SS1

ROUTINE:

Expand the Controller Organizer and select the routine **FuncRoutine**.

The blank editor with Sheet 1 will be displayed.



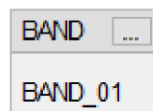
FUNCTION BLOCK ROUTINE

[See Function Block Assist for program.](#)

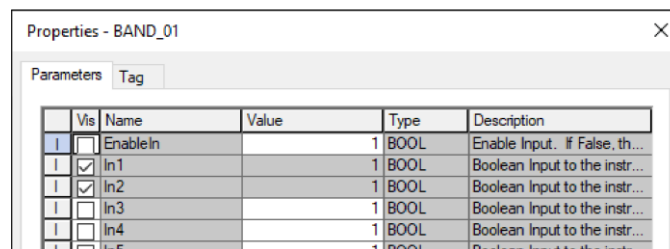
Add the following instructions.

Instruction Tabs:	IREF -	Favorites
	OREF -	Favorites
	BAND -	Favorites or Move/Logical
	BOR -	Favorites or Move/Logical
	BNOT -	Favorites or Move/Logical

Parameters for BAND_01, select the Properties Button.



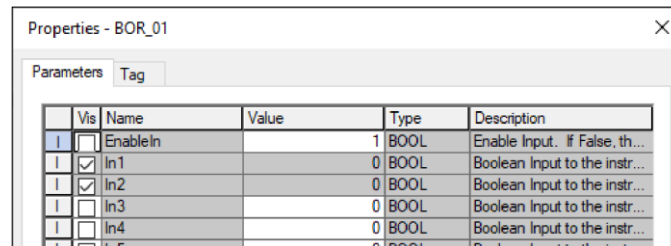
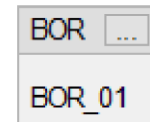
Uncheck the **In3** and **In4** parameters.



continue

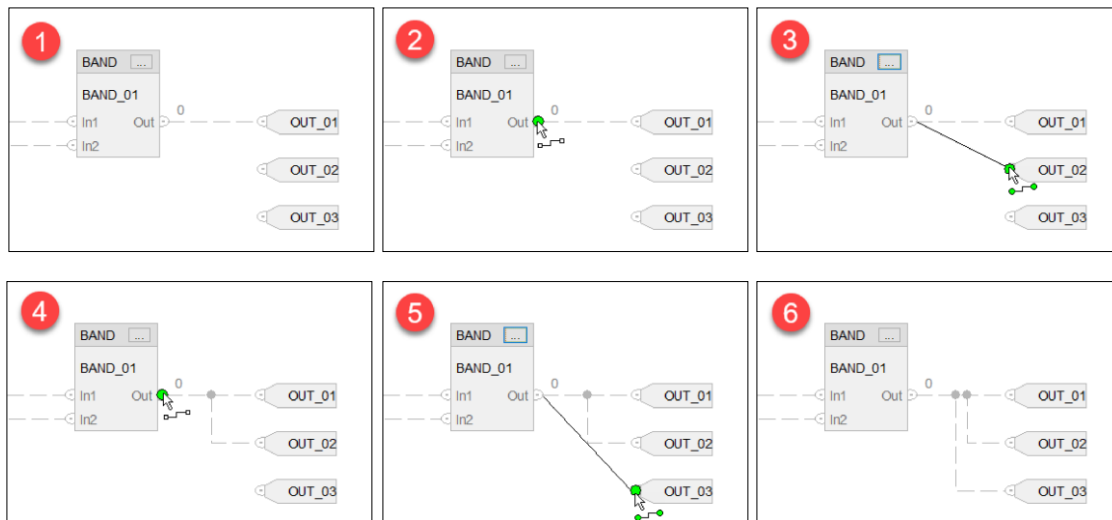
Parameters for BOR_01, select the Properties Button.

Uncheck the **In3** and **In4** parameters.



CONNECTORS

When adding Multiple Connections to the **BAND_01** instruction **Out** “pin”, drag each connector to the BAND instruction’s Out “pin”. The editor will move the connection point.

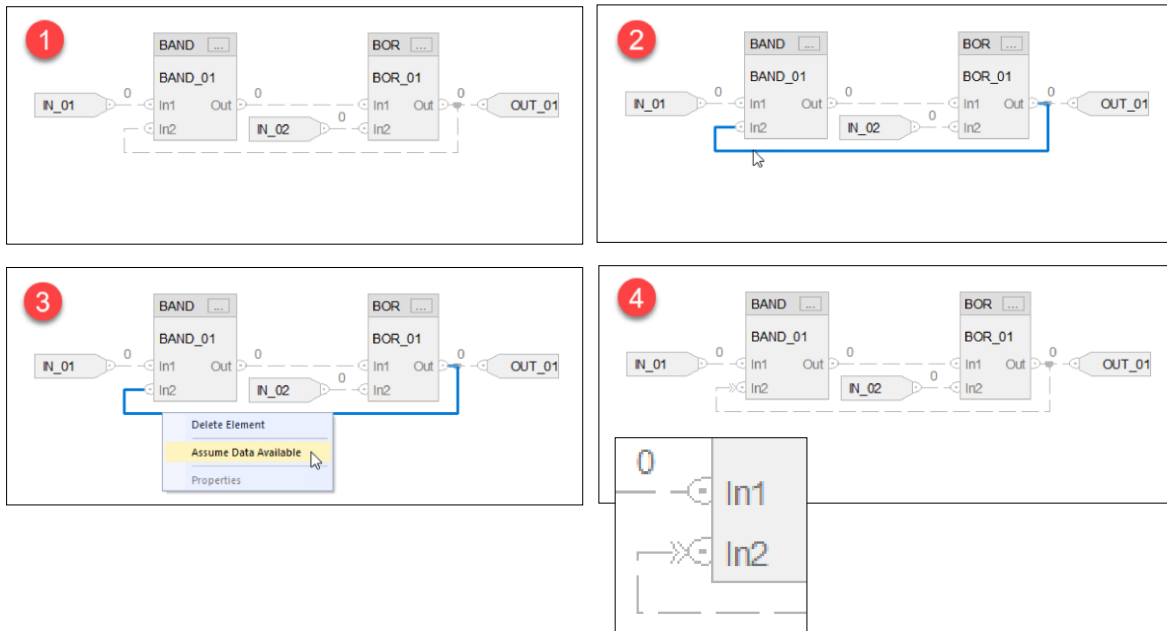


Note: The placement of the instructions will change how the connection lines are drawn by the editor.

continue

ASSUMED DATA on BOR_1, In2 PIN

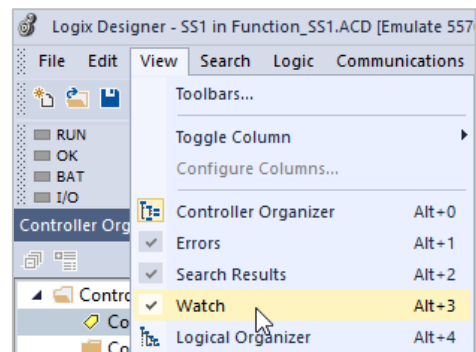
After drawing the connector, right click on the line, from the context menu select **Assume Data Available**. A **double arrowhead** shows Assumed Data.



WATCH WINDOW AND QUICK WATCH

OPEN WATCH WINDOW

From the Menu Toolbar, select **View** and select **Watch**.

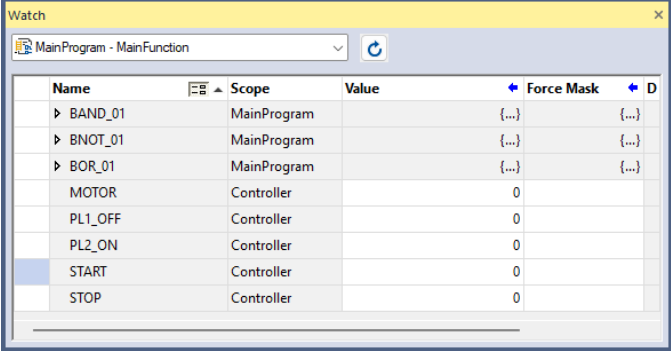


Notice: The tabs at the bottom of the Watch Window. The same dialog window is shared, with the verify **Errors** and **Search** results. Select the **Watch** tab to view tags.

continue



By default, the tags in the current routine are displayed.

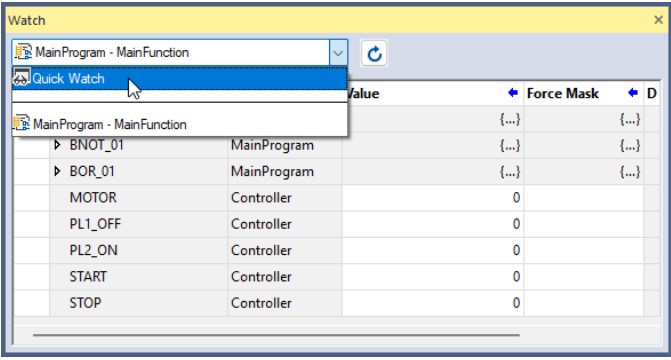


The screenshot shows the 'Watch' window with a dropdown menu set to 'MainProgram - MainFunction'. The table below lists the variables and their current values.

Name	Scope	Value	Force Mask	D
▶ BAND_01	MainProgram	{...}	{...}	{...}
▶ BNOT_01	MainProgram	{...}	{...}	{...}
▶ BOR_01	MainProgram	{...}	{...}	{...}
MOTOR	Controller	0		
PL1_OFF	Controller	0		
PL2_ON	Controller	0		
START	Controller	0		
STOP	Controller	0		

CREATE A QUICK WATCH

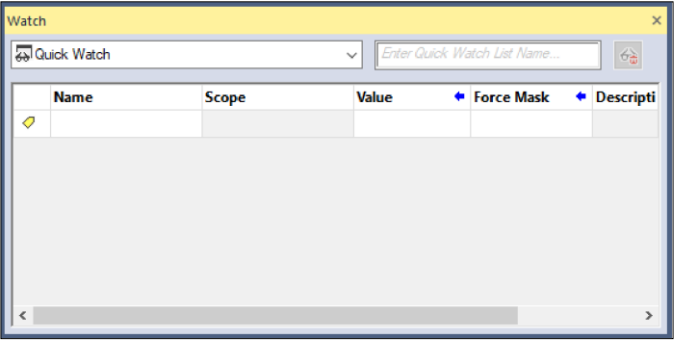
From the dropdown select **Quick Watch**.



The screenshot shows the 'Watch' window with the dropdown menu open and 'Quick Watch' selected. The table below lists the variables and their current values.

Name	Scope	Value	Force Mask	D
▶ BNOT_01	MainProgram	{...}	{...}	{...}
▶ BOR_01	MainProgram	{...}	{...}	{...}
MOTOR	Controller	0		
PL1_OFF	Controller	0		
PL2_ON	Controller	0		
START	Controller	0		
STOP	Controller	0		

A blank Watch Window is opened.

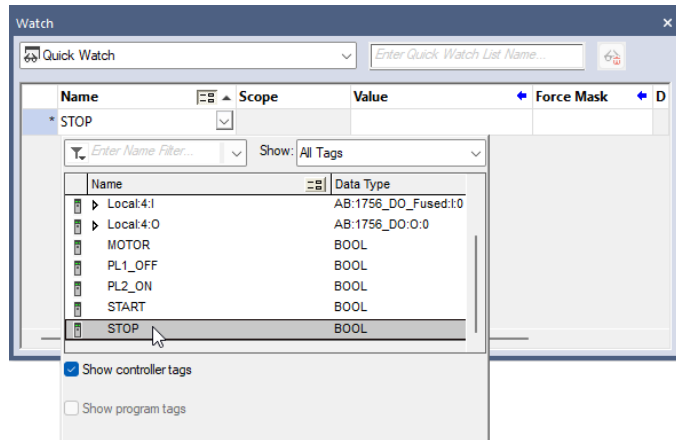


The screenshot shows a blank 'Watch' window with the dropdown menu set to 'Quick Watch'. The table below lists the variables and their current values.

Name	Scope	Value	Force Mask	Descripti
------	-------	-------	------------	-----------

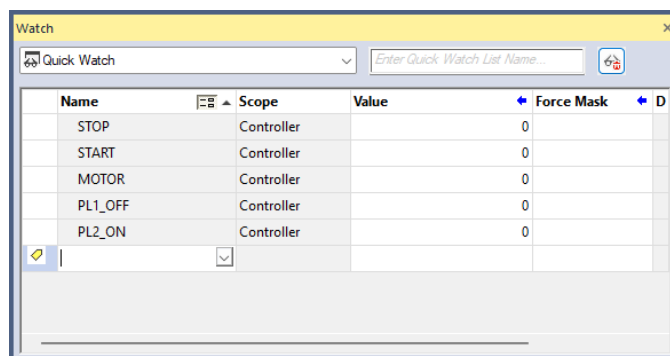
continue

To select tags, left click on the **Name** field dropdown arrow and select the tag from the listing.



NOTE: THE TAGS LIST ARE THE SAME AS THE “CURRENT ROUTINE” YOU MUST RECREATE THE SAME TAG LISTING AS A NAMED QUICK WATCH FOR CREDIT.

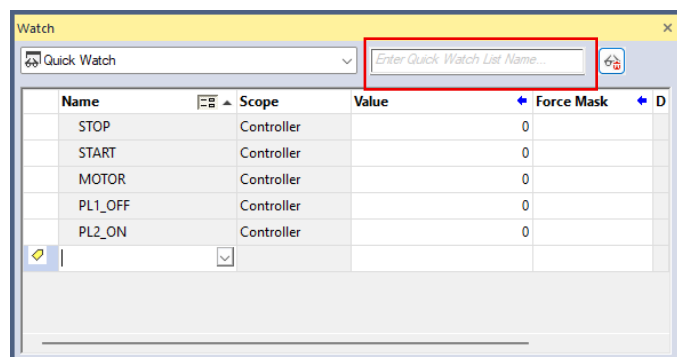
Completed Quick Watch.



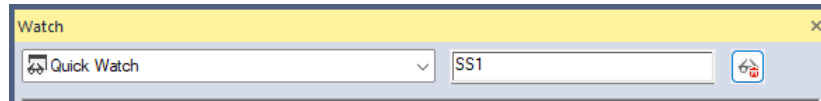
SAVING A QUICK WATCH

To save the Quick Watch...

Select the *Enter Quick Watch List Name...* field.

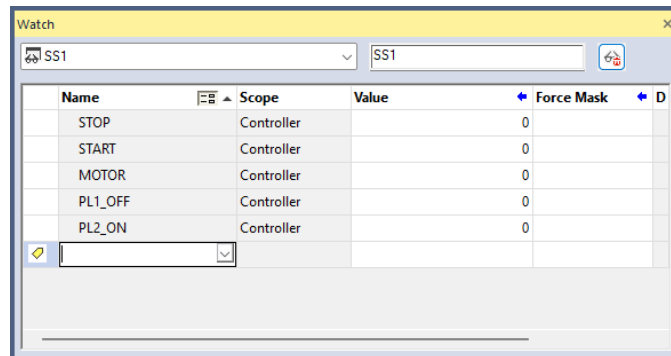


Type in a name for the current quick watch window and press **Enter**.



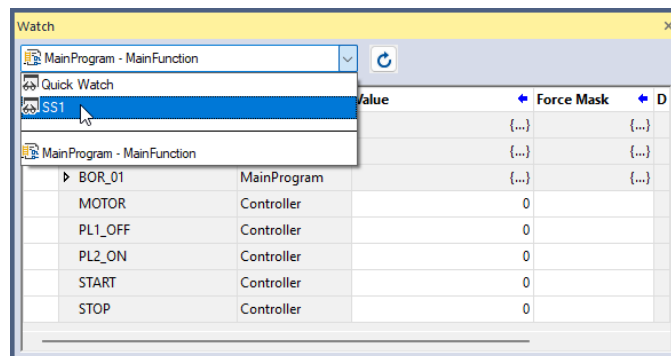
The tags are saved under the named window.

Note: Multiple named Watch Windows of different tags may be saved.



TO SELECT A SAVED QUICK WATCH WINDOW

In the opened a saved Watch Window, click on the down arrow and select the named watch from the list.



DOWNLOAD

Download the PLC program and place the PLC into RUN

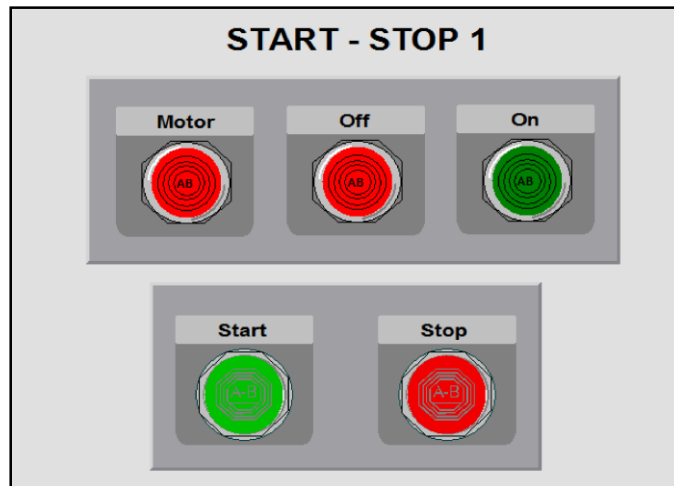
An HMI has been created to allow you to test the logic, SS1_HMI.

The HMI is a runtime file.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

continue

TESTING



*Notice the tag **Value** STOP pushbutton is set to 1.*

*Notice the tag **Value** PL1_OFF indicator is set to 1.*

1. On the HMI, left click and hold the **START** pushbutton.

Value field:

- The START pushbutton is set to **1**.
- The PL1_OFF indicator will turn Off, reset to **0**.
- MOTOR will turn On, set to **1**.
- The PL2_ON will turn On, set to **1**.
- Release the START pushbutton.
- The **Value** changes to **0**.

continue



2. Left click on the **STOP** pushbutton.

Value field:

- STOP changes to **0** *to emulate the pressing of the STOP pushbutton.*
- The PL1_OFF indicator will turn On, set to **1**.
- MOTOR will turn Off, reset to **0**.
- The PL2_ON will turn Off, reset to **0**.
- Release the STOP pushbutton.
- The **Value** changes to **0**.



LAB: WASHER STATION TIMER**PROCESS MODEL:** WASH_TON**OBJECTIVE:** Create function block program using the timer on instruction.

Create a program that when a momentary normally open pushbutton is pressed and held, two clamp solenoids will be energized. A normally open limit switch for each clamp will be set when that clamp is in position. When the two clamps are in position, a spray wash will clean the part for ten seconds. Releasing the pushbutton will turn off the spray wash and clamps.

INPUT:

Address	Controller Tag	Descriptions
Local:3:I.Data.0	PUSHBUTTON	Pushbutton Wash Cycle
Local:3:I.Data.1	SWITCH_CLAMP1	Limit Switch Clamp 1
Local:3:I.Data.2	SWITCH_CLAMP2	Limit Switch Clamp 2

OUTPUT:

Address	Controller Tag	Descriptions
Local:4:O.Data.0	SOLENOID_CLAMP1	Clamp 1 Solenoid
Local:4:O.Data.1	SOLENOID_CLAMP2	Clamp 2 Solenoid
Local:4:O.Data.2	WASH_VALVE	Spray Wash Valve

FUNCTION BLOCKS:*Function Block Tags are created as the instructions are added.*

Param/Local Tag	Descriptions
BAND_01	AND for Pushbutton and Clamp Switches
BAND_02	AND for Pushbutton and Clamp Switches and Timer
BNOT_01	Invert Timer Done
WASH_TIMER	Wash Valve Timer
	<i>Note: Rename the Timer TONR_1 after dragging into editor, either rename in Param/Local Tag Table or instruction's Properties.</i>



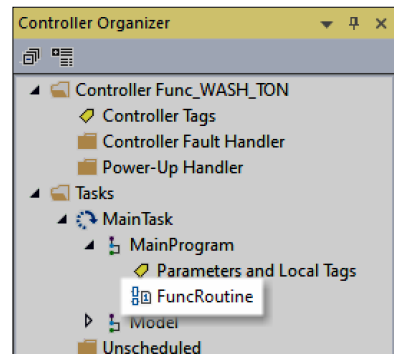
PROGRAM LOCATION:

Studio 5000 Folder
Course Day/Time Folder
Languages Folder
Function Block Folder

FILE NAME: Function_WASH_TON

ROUTINE:

Expand the Controller Organizer and select the routine **FuncRoutine**.



The blank editor with Sheet 1 will be displayed.

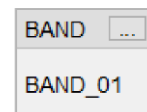
FUNCTION BLOCK ROUTINE

[See Function Block Assist for program.](#)

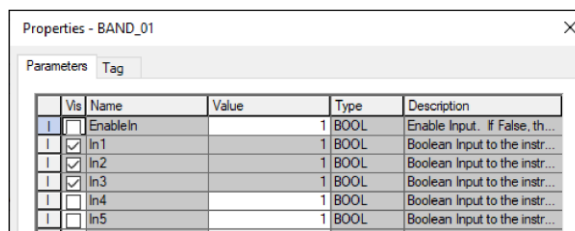
Add the following instructions.

Instruction Tabs:	IREF -	Favorites
	OREF -	Favorites
	BAND -	Favorites or Move/Logical
	BNOT -	Favorites or Move/Logical
	TONR -	Timer/Counter

Parameters for BAND_1, select the Properties Button.



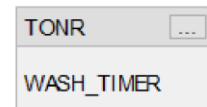
Uncheck the **In4** parameter.



continue



Parameters for TONR WASH_TIMER, select the Properties Button.



NOTICE: The renamed Timer instruction.

If you have not renamed from TONR_1 do so now, either rename in the Controller Properties **Param/Local Tag Table** or instruction's **Properties Tag** tab.

Uncheck the parameters for: **PRE**
Reset
ACC

A screenshot of a software window titled 'Properties - WASH_TIMER'. It has two tabs: 'Parameters' and 'Tag'. The 'Parameters' tab is active, showing a table with columns: 'Vis', 'Name', 'Value', 'Type', and 'Description'.

Vis	Name	Value	Type	Description
☐	EnableIn		1 BOOL	Enable Input. If False, th...
☒	TimerEnable		0 BOOL	Timer Enable. Enable tim...
☐	PRE	10000	DINT	PRE. Timer Preset Value...
☐	Reset		0 BOOL	Reset. Request to reset t...
☐	EnableOut		0 BOOL	Enable Output.
☐	ACC		0 DINT	Accumulated time in millis...
☐	EN		0 BOOL	EN. Timer Enabled output...
☐	TT		0 BOOL	TT. Timer Timing output. ...
☒	DN		0 BOOL	DN. Timing Done output...
☐	Status	15#0000_0000	DINT	Rt mapped status of the f...

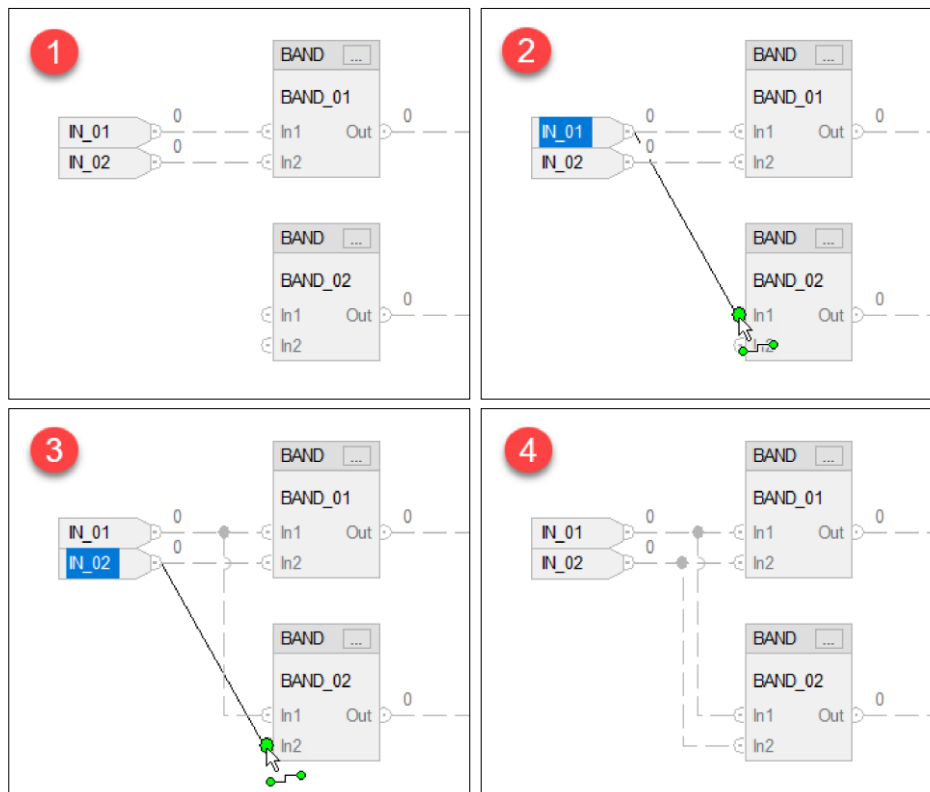
At **PRE** enter 10000 for the **Value**.

continue for connectors

CONNECTORS

When adding multiple connections to the same instruction “pin”, select the “pin” and drag to the other instruction’s “pin”. The editor will move the connection point.

Note: The placement of the instructions will change how the connection lines are drawn by the editor.



continue

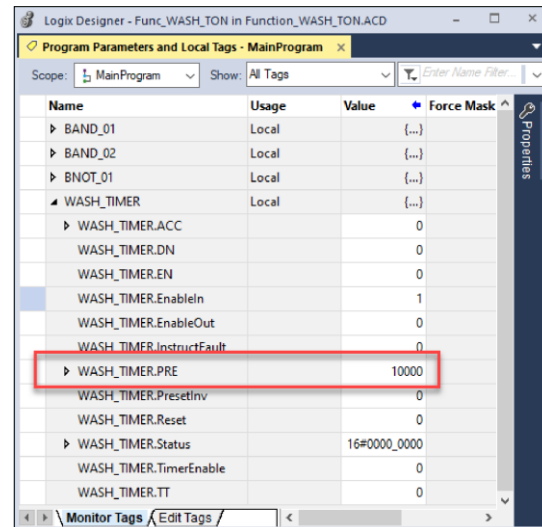


ALTERNATIVE: SETTING TIMER PRESET

In the Controller Organizer, select the MainProgram's Param/Local Tags.

Select the **Monitor Tags** tab.

Expand **WASH_TIMER** to view **WASH_TIMER.PRE** and add in the value of 10000.

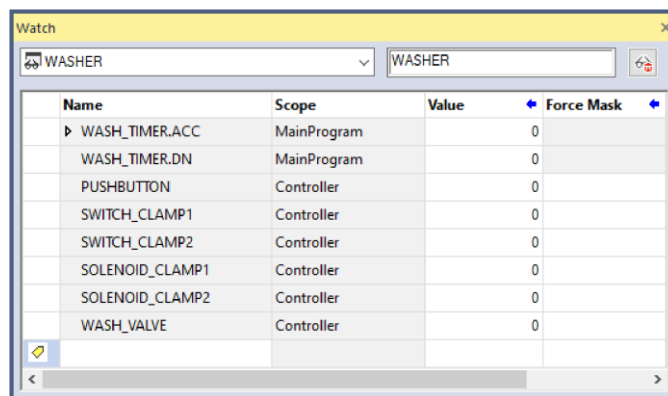


WATCH WINDOW AND QUICK WATCH

See SS1 for similar details on Watch Window creation and usage, pages 4 to 7.

Completed Quick Watch.

Name: WASHER



continue

DOWNLOAD

Download the PLC program and place the PLC into RUN

An HMI has been created to allow you to test the logic, WASH_TON_HMI.

The HMI is a runtime file.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

TESTING



1. On the HMI, left click and hold the **PUSHBUTTON**.

Value field:

- PUSHBUTTON changes to the value to **1**.
- The SOLENOID_CLAMP1 and SOLENOID_CLAMP2 will turn On, set to **1**.
- CLAMP_SWITCH1 and CLAMP_SWITCH2 close, set to **1**.
- The WASH_VALVE turns On, set to **1**.
- The timer begins to elapse.

continue

- When the timer's preset and accumulated are equal, the Done bit is set.
- The Wash Valve turns Off, reset to **0**.

2. Release the **PUSHBUTTON**.

Value field:

- Changes PUSHBUTTON value to **0**.
- Turns Off the SOLENOIDS_CLAMP1 and SOLENOIDS_CLAMP1, all reset **0**.
- Opens the SWITCH_CLAMP1 and SWITCH_CLAMP2, all reset **0**.
- The WASH_VALVE turns Off, reset to **0**.
- Timer is reset.



LAB: TIMED PROCESS OUTPUT
PROGRAM FILE: PROCESS_OUTPUT
OBJECTIVE: Function and arithmetic instructions.

Create the program for a process that uses multi-layers of a sheet material. The type and number of sheets loaded depends on the order. The operator enters one of four types of sheet materials that will be used and the number of sheet layers. Only one type of material is used at a time, and up to six sheets may be used.

The operator presses a HMI panel pushbutton that corresponds to the material type, then presses a HMI panel pushbutton for the number of sheets. When the operator presses and releases the START pushbutton, a process output will turn on for the computing determined time value based on the type and number of sheets. The following parameters are used to then control the process timer.

Material A - multiplier of 0.73 times the number of sheets
 Material B - multiplier of 1.05 times the number of sheets
 Material C - multiplier of 1.23 times the number of sheets
 Material D - multiplier of 1.86 times the number of sheets

INPUTS:

Address	Controller Tag	Descriptions
Local:3:I.Data.0	SW_START	START Momentary N. O. Pushbutton

OUTPUT:

Address	Controller Tag	Descriptions
Local:4:O.Data.0	PROCESS_OUTPUT	Process Output

INTERNALS:

Param/Local Tag	Data Type	Descriptions
A_MATERIAL_MULTIPLIER	Real	Material A Multiplier
B_MATERIAL_MULTIPLIER	Real	Material B Multiplier
C_MATERIAL_MULTIPLIER	Real	Material C Multiplier
D_MATERIAL_MULTIPLIER	Real	Material D Multiplier
MATERIAL_MULTI_VALUE	Real	Material Multiplier Value



HMI_NUM_SHEET	Integer	Number of Material Sheets
HMI_TYPE_SHEET	Integer	Type of Sheets

FUNCTION BLOCKS:

Function Block Tags are created as the instructions are added.

Param/Local Tag	Descriptions
MUX_01	Select Material Type Process Value
MUL_01	Determine Process Value
MUL_02	Convert to Milliseconds
BOR_01	AND for Pushbutton and Clamp Switches
BAND_01	AND for Pushbutton and Clamp Switches and Timer
TONR_01	Process Timer
BNOT_01	Invert Timer Done

PROGRAM LOCATION:

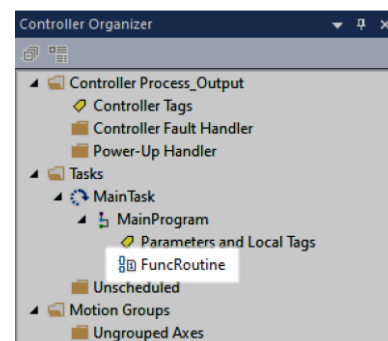
Studio 5000 Folder
Course Day/Time Folder
Languages Folder
Function Block Folder

FILE NAME: Function_PROCESS_OUTPUT

ROUTINE:

Expand the Controller Organizer and select the routine **FuncRoutine**.

The blank editor with Sheet 1 will be displayed.



continue

FUNCTION BLOCK ROUTINE

See Function Block Assist for program.

ADD ROUTINE

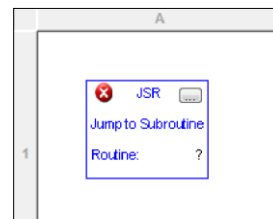
File Name: FuncHMI

Program Type: Function Block

ROUTINE FuncRoutine (Main Routine)

Add the following instructions to **Sheet 1**.

Instruction Tabs: JSR - Program Control



Create Sheet 2 and add the following instructions.



New Sheet icon.

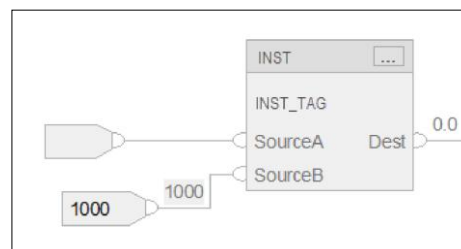


Instruction Tabs: IREF - Favorites
OREF - Favorites
MUL - Compute/Math
OCON - Favorites

Add the logic to compute from the Number of Sheets and the Material Multiplier Value and then change the time value to milliseconds.

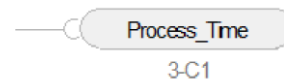
Example of using a CONSTANT as an In Parameter.

- Drag an IREF into the logic.
- Enter the Constant value.
- Draw connector to pin.



continue

The resulting product Destination in milliseconds is assigned to an OCON labeled **Process_Time**.

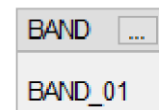


*The label Process_Tame is **not** a Tag. The label is a placeholder for the value.*

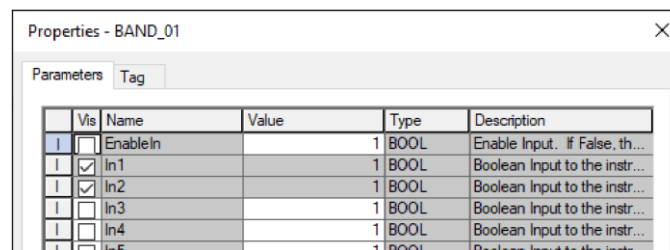
Create Sheet 3 and add the following instructions.

Instruction Tabs: BAND -	Favorites or Move/Logical
BNOT -	Favorites or Move/Logical
TONR -	Timer/Counter
ICON -	Favorites

Parameters for BAND_01, select the Properties Button.

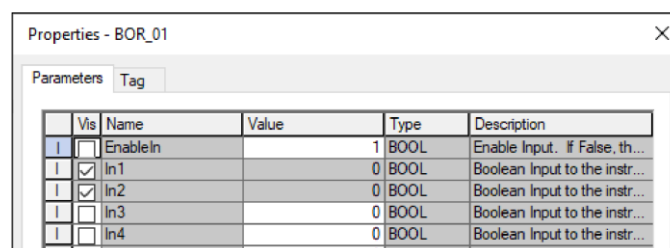


Uncheck the **In3** and **In4** parameters.



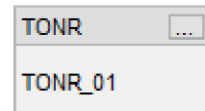
Parameters for BOR_01, select the Properties Button.

Uncheck the **In3** and **In4** parameters.

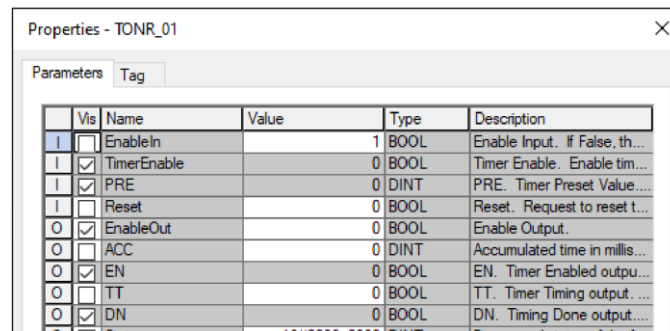


continue

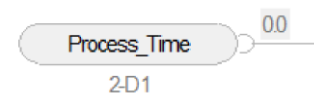
Parameters for TONR_01, select the Properties Button.



Uncheck the parameters for: **Reset**
ACC



The ICON labeled **Process_Time** is used to transfer from Sheet 2 the millisecond value to the timer's Preset.



ADD ROUTINE

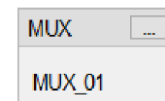
File Name: FuncHMI
Program Type: Function Block

Add the following instructions.

Instruction Tabs:

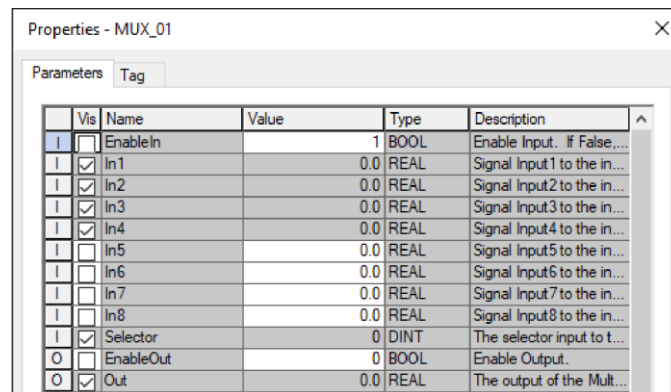
IREF - Favorites
OREF - Favorites
MUX - Select/Limits

Parameters for MUX_01, select the Properties Button.



continue for parameters illustration

Uncheck the **In5**, **In6**, **In7**, and **In8** parameters.

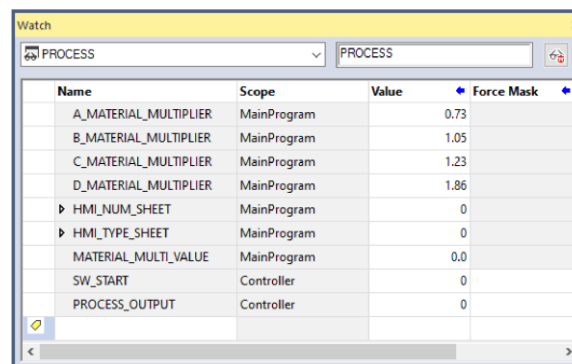


WATCH WINDOW AND QUICK WATCH

See SS1 for similar details on Watch Window creation and usage, pages 4 to 7.

Completed Quick Watch.

Name: PROCESS



SETTING PROCESS VALUE FOR MATERIAL TYPE

Open the named Quick Watch Window.

Set the Values for **A_** through **D_MATERIAL_MULTIPLIER** from the Quick Watch Window.

Save the program.

continue



DOWNLOAD

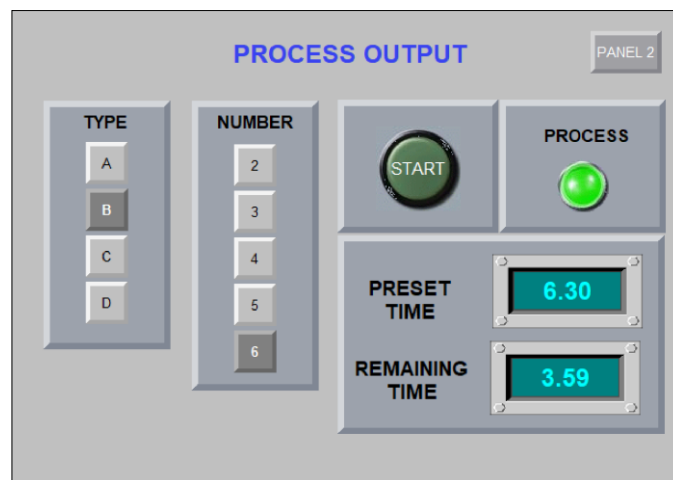
Download the PLC program and place the PLC into RUN

*An HMI has been created to allow you to test the logic, **PROCESS_OUTPUT_HMI**.*

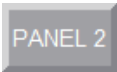
The HMI is a runtime file.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

TESTING



ENTERING MATERIAL MULTIPLIER

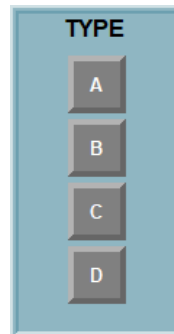
1. Press the button to go to **PANEL 2**. 
2. Enter the values for the **A, B, C, and D MATERIALS**.
3. Press the button to return to the **MAIN** screen 

MULTIPLIER VALUE SET-UP	
MATERIAL A	0.73
MATERIAL B	1.05
MATERIAL C	1.23
MATERIAL D	1.86

continue

SELECTING SHEET TYPE AND NUMBER

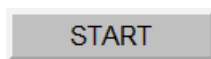
1. Press the Button for **TYPE B**.



2. Press the Button for **NUMBER 2**.

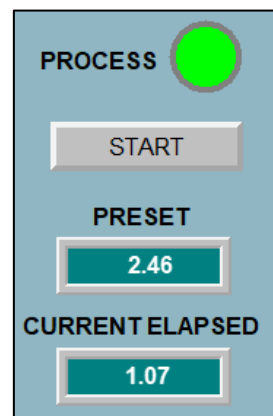


3. Press the **START** button.



Value field:

- SW_START change to the value to 1.
- Release the START pushbutton.
- The **Value** changes to **0** to emulate the releasing of the Start pushbutton.
- The MATERIAL_MULTI_VALUE and OP_NUM_SHEET will be multiplied.
- The product of the previous multiply instruction will multiply by 1000 to convert to milliseconds.



continue



- The product is provided to the timer instruction's preset (PRE) value.
 - The timer begins to elapse.
 - The PROCESS_OUTPUT turns On, set to **1**.
 - When the timer's preset and accumulated are equal, the Done bit is set.
 - The PROCESS_OUTPUT Off, set to **0**.
4. Select the other **TYPE** and **NUMBER** buttons and then **START** to verify the different values calculated values for the Timer's **PRESET**.

